Unilateral subcostal anterior quadratus lumborum block versus intrathecal morphine for postoperative pain in laparoscopic nephrectomy: a randomized controlled trial

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► Additional supplemental material is published online only. To view, please visit the journal online (https://doi.org/10.1136/rapm-2025-106844).

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Received 17 May 2025
Accepted 23 June 2025

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To cite: Kaya C, Dost B, Turunc E, *et al*. *Reg Anesth Pain Med* Epub ahead of print: [*please include* Day Month Year]. doi:10.1136/rapm-2025-106844

ABSTRACT

Background Intrathecal morphine (ITM) offers effective analgesia after laparoscopic nephrectomy but may cause side effects. The anterior subcostal quadratus lumborum (QL) block has been proposed as a safer alternative. This study evaluated whether anterior subcostal QL block is non-inferior to ITM in reducing postoperative opioid use.

Methods This single-center, randomized, observer-blinded, non-inferiority trial allocated 80 adult patients undergoing elective laparoscopic nephrectomy to receive single-injection unilateral anterior subcostal QL block (0.25% bupivacaine (0.4 mL/kg) containing epinephrine 1:400,000) or spinal anesthesia with 5 mcg/kg ITM (maximum 200 mcg) plus 7.5 mg isobaric bupivacaine. The primary outcome was the 24-hour cumulative intravenous morphine milligram equivalent (MME). Pain scores were recorded at rest and during activity at 0, 3, 6, 12, and 24 hours postoperatively. Patient satisfaction, recovery quality, and adverse effects were also evaluated.

Results The median difference in 24 hours intravenous MME consumption between anterior subcostal QL block and ITM was 9.5 mg (95% CI 7 to 12; p<0.001), exceeding the non-inferiority margin. Patients with ITM reported lower pain scores and greater satisfaction at 24 hours (median (Q1–Q3), ITM 108 (102.25–113.5) vs anterior subcostal QL block 75.5 (58–100)) and discharge (ITM 117 (115–123.75) vs anterior subcostal QL block 94 (77–110); p<0.001). However, no between-group differences were observed in 30-day complication rates, including the Clavien–Dindo and Comprehensive Complication Index scores. Adverse effects, such as nausea, pruritus, and respiratory events, were comparable.

Conclusions Although designed as a non-inferiority trial, anterior subcostal QL block did not meet the threshold. While ITM resulted in lower opioid use, the study was not powered to establish superiority. Anterior subcostal QL block may be considered for patients at risk of ITM-related respiratory depression; however, its inconsistent cranial spread and limited visceral analgesia may explain its lower efficacy. Caution is also required in anticoagulated patients because of the deep anatomical location of the block.

Trial registration number ClinicalTrials.gov: NCT06630858.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Intrathecal morphine (ITM) provides effective and long-lasting analgesia after laparoscopic nephrectomy, including visceral pain relief, but has dose-related side effects. The anterior subcostal quadratus lumborum (QL) block has been proposed as a safer ultrasound-guided alternative.

WHAT THIS STUDY ADDS

⇒ Although a non-inferiority design was used, the 24 hours cumulative intravenous morphine milligram equivalent difference exceeded the non-inferiority margin, indicating that the anterior subcostal QL block was inferior and that ITM provided statistically and clinically better analgesia; however, the study was not designed for direct comparisons beyond non-inferiority.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE, OR POLICY

⇒ These findings prompt reconsideration of anterior subcostal QL block use, especially for procedures involving visceral pain. While newer interfascial blocks (eg, external oblique intercostal plane blocks) show promise, they should be validated through well-designed comparative trials before routine adoption.

INTRODUCTION

Laparoscopic nephrectomy is associated with significant postoperative pain owing to visceral manipulation and abdominal wall trauma.[1] Intrathecal morphine (ITM) effectively manages somatic and visceral postoperative pain, offering prolonged analgesia. However, its use is limited by dose-dependent adverse effects, such as pruritus, urinary retention, sedation, and respiratory depression.[2–4] ITM provides prolonged analgesia by activating spinal μ-opioid receptors due to its hydrophilic nature and slow clearance from cerebrospinal fluid. However, this also increases the risk of delayed respiratory depression from rostral spread to the brainstem, usually occurring 6–12 hours after injection. Thus,

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extended monitoring for up to 24 hours is recommended. Still, recent data suggest that low-dose ITM (≤200 µg) can be safely used with standard ward monitoring in selected patients.[2 3]

Ultrasound-guided quadratus lumborum (QL) block is a recent truncal interfascial block that provides effective analgesia for abdominal surgeries.[5] The anterior subcostal approach involves injecting local anesthetic between the QL muscle fascia and the anterior thoracolumbar fascia at the L1/L2 level.[6] Initial cadaveric and clinical studies suggest that anterior subcostal QL blocks might provide thoracic dermatomal coverage through cranial spread to the paravertebral space.[6 7] However, more recent evidence indicates that such spread is inconsistent, and the analgesic effect may primarily result from thoracolumbar fascia involvement.[7–9]

Our previous study comparing the anterior subcostal QL block with the thoracic paravertebral block (TPVB) in laparoscopic nephrectomy showed a modest but statistically significant reduction in 24-hour morphine consumption in the TPVB group, suggesting comparable analgesic efficacy.[10] This finding is consistent with the modest opioid-sparing effects reported for the anterior subcostal QL block in the literature and supports its further evaluation of ITM. Given the risks of ITM, exploring alternative analgesic techniques with comparable efficacy and fewer adverse effects is important. This randomized study tested whether the anterior subcostal QL block is non-inferior to ITM in reducing postoperative opioid consumption after laparoscopic nephrectomy.

METHODS

Trial design

This prospective, randomized, controlled, observer-blinded, non-inferiority trial was approved by the Ondokuz Mayis University Clinical Trials Ethics Committee (approval no. 2023/212) on 17 September 2024 and registered in ClinicalTrials.gov (ID: NCT06630858; principal investigator: CK; registration date, 7 October 2024). The study adhered to Good Clinical Practice standards and the Declaration of Helsinki and followed the Consolidated Standards of Reporting Trials 2010 guidelines for non-inferiority and equivalence trials.[11 12]

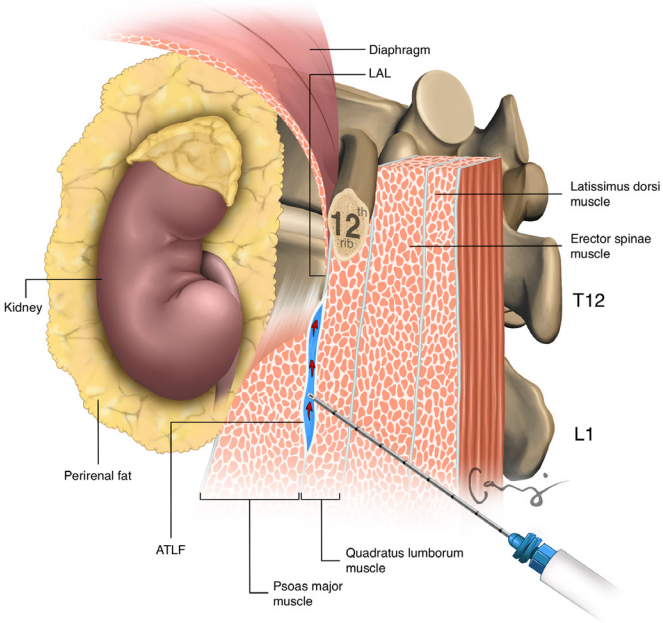

Figure 1 The paramedian schematic diagram illustrates the needle trajectory for the anterior subcostal QL block and the cephalad spread of the LA. Key anatomical structures, including the latissimus dorsi, erector spinae, QL, and psoas major muscles, as well as the 12th rib and T12 vertebra, are shown. The LA spreads posterior to the LAL and reaches the paravertebral space. ATLF, anterior thoracolumbar fascia; LA, local anesthetic; LAL, lateral arcuate ligament; QL, quadratus lumborum.

Study participants

Patients scheduled for elective laparoscopic unilateral nephrectomy at Ondokuz Mayis University Hospital between October 2024 and April 2025 were screened for eligibility. Participants aged 18–70 years with an American Society of Anesthesiologists (ASA) physical status of I–III, who provided informed consent, were included. The exclusion criteria included refusal to participate; body mass index >35 kg/m^2; contraindications to peripheral nerve or spinal anesthesia (eg, coagulopathy, infection); anatomical barriers to regional anesthesia; significant cardiovascular, cerebrovascular, or major organ dysfunction; drug allergies; history of substance abuse; chronic opioid use within the past 3 months (≥15 mg/day for≥30 days);[13] psychiatric illness; or communication disorders.

Randomization, allocation concealment, and blinding

Three physicians (CK, BD, and ET) assessed participant eligibility and obtained written informed consent before enrollment. Patients were randomized to receive either a unilateral anterior subcostal QL block (n=40) or ITM (n=40). The randomization was conducted by an independent physician using the "Research Randomizer" tool (Urbaniak and Plous, 2013), applying a 1:1 ratio and a permuted block size of 2. The allocation was concealed in sequentially numbered opaque envelopes, stored securely by the principal investigator (CK) until the day of surgery, and opened by the administering anesthesiologist shortly before the intervention. All procedures were performed by faculty anesthesiologists skilled in ultrasound-guided nerve blocks and spinal anesthesia. An independently trained anesthesiologist evaluated the clinical features of the blocks. Because both the anterior subcostal QL block and ITM were administered before induction, patient blinding was not feasible. Spinal injections were performed in the operating room, whereas the anterior subcostal QL block was performed in a separate block room. All other investigators, care providers, outcome assessors, statisticians, and surgeons were blinded to the group allocation.

Interventions

Participants were placed in a block room or operating theater. Following intravenous cannula placement and application of standard ASA monitoring (electrocardiography, noninvasive blood pressure, and peripheral oxygen saturation (SpO2)), all patients received intravenous midazolam at 0.02 mg/kg (calculated based on ideal body weight (IBW)) to achieve a Ramsay sedation score of 2. No opioids or additional sedatives were administered during block placement. Supplemental oxygen (2–3 L/min) was provided via nasal prongs. The anterior subcostal QL block was performed 30 min preoperatively, and the ITM was administered before the induction of general anesthesia.

Anterior subcostal quadratus lumborum block

We performed a unilateral anterior subcostal QL block as described by Elsharkawy *et al*.[6] The ultrasound probe was positioned posteriorly at the L1–L2 level in the parasagittal oblique plane, allowing visualization of the erector spinae, QL, and psoas major muscles. Using an in-plane technique, the needle was advanced into the interfascial

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plane between the QL and psoas major muscles (figure 1, online supplemental appendix 1). Following an injection of 0.25% bupivacaine (Marcaine, AstraZeneca, Türkiye) with 1:400 000 epinephrine at 0.4 mL/kg (IBW), local anesthetic spread cranially between the QL and psoas major muscles toward the 12th rib. Sensory block assessments were performed 30 min after the block by an experienced anesthesiologist who was not part of the study team and was blinded to the group allocation. Pinprick testing was conducted across dermatomes T7–L1 using a standardized 3-point scale (0 no change, 1 touch without pain, 2 no sensation). Block success was defined as a score of ≥1 at all dermatomes on the surgical side. Even a single dermatome with a score of 0 was considered block failure.

Intrathecal morphine administration

In the ITM group, patients received an intrathecal injection in the sitting position using a 25-gauge spinal needle (Pencan, B. Braun, Germany) at the L3–L4 or L4–L5 levels via a midline or paramedian approach. Two hundred micrograms of ITM is suggested in the literature as an effective dose, balancing analgesic efficacy and side effects.[2] Accordingly, preservative-free morphine was administered at a dose of 5 mcg/kg (maximum 200 mcg), combined with 7.5 mg of isobaric bupivacaine, by the attending anesthesiologist. This dosing approach reflects institutional practice and aligns with the Enhanced Recovery After Surgery and Procedure-Specific Postoperative Pain Management guidelines. Low-dose ITM (50–200 mcg) provides up to 48 hours of analgesia with minimal risk of respiratory depression.[2] Isobaric bupivacaine ensures the rapid sensory block onset, while morphine provides prolonged analgesia. The combination enhances pain control with fewer side effects.[14]

ITM was administered in a separate preparation area shortly before the patient was transferred to the operating room for general anesthesia induction. The anterior subcostal QL block was performed approximately 30 min before surgery in a dedicated block room. Twenty-four-hour intravenous morphine milligram equivalent (MME) data included all postoperative opioid consumptions via a patient-controlled analgesia (PCA) device and rescue opioids in the post-anesthesia care unit (PACU) and ward. ITM was excluded from the intravenous MME calculations to ensure a valid comparison of systemic opioid use.

Anesthesia and analgesia management

A detailed description of the anesthetic and multimodal pain management is provided in online supplemental appendix 2.

Postoperative nausea and vomiting

Postoperative nausea and vomiting (PONV) was managed according to institutional protocols. All patients received 8 mg intravenous dexamethasone before induction and 0.15 mg/kg (IBW) intravenous ondansetron (maximum 8 mg) 20 min before the end of surgery. PONV severity was assessed using a 5-point scale (0=no nausea, 1=slight nausea, 2=moderate nausea, 3=vomiting once, and 4=vomiting more than once). In cases with a score of ≥3, repeat ondansetron (4 mg intravenous) was permitted, although infrequently used. If symptoms persisted, 10 mg intravenous metoclopramide diluted in 100 mL saline was administered via slow infusion. All PONV scores and antiemetic treatments were documented for 24 hours postoperatively.

Quality of recovery

Recovery quality was assessed using the Turkish version of the 15-item quality of recovery (QoR-15) score. QoR-15 is a patient-reported tool that assesses postoperative functional recovery and well-being across five domains: physical comfort, pain, physical independence, psychological support, and emotional state. Scores range from 0 (poor recovery) to 150 (excellent recovery) and are evaluated preoperatively, at 24 hours postoperatively, and at discharge.[15 16]

Postoperative respiratory depression

Respiratory depression was monitored using both SpO2 and respiratory rate (RR) during the first 24 postoperative h in accordance with a standardized protocol applied uniformly to all patients, regardless of group allocation. SpO2 was continuously monitored via pulse oximetry, and RR via capnography through a nasal cannula delivering 2 L/min of oxygen, enabling real-time respiratory assessment. Respiratory depression was defined as oxygen saturation <90% or RR <8 breaths per minute. In cases of respiratory depression, airway patency was maintained, and 5 L/min of oxygen was administered via a face mask. Naloxone was administered if opioid-induced respiratory depression was suspected, and assisted ventilation was provided if required. The incidence of respiratory depression was also recorded.

Postoperative monitoring and outcomes assessment

All patients were monitored for 30 days postoperatively, with complications classified using the Clavien–Dindo system.[17] Morbidity was assessed using the Comprehensive Complication Index.[18] The initial health condition and mortality risk were assessed using the Charlson Comorbidity Index.

Outcomes

Primary outcome

The 24 hours cumulative intravenous MME consumption was calculated from when patients regained consciousness and could communicate in the PACU. It only included postoperative opioid administration delivered via PCA or rescue analgesia in the ward. The intraoperative opioid doses were not included in the calculations.

Secondary outcomes

Secondary outcomes included numeric rating scale (NRS) scores at rest and during activity (coughing) in the PACU and at 3, 6, 12, and 24 hours; 12 hours intravenous MME consumption; time to first PCA bolus; QoR-15 scores; intraoperative variables including heart rate, mean arterial pressure, and remifentanil consumption, recorded at predefined time points; rescue analgesia and antiemetic requirements; sedation levels (0–3 scale); presence of pruritus (≥1, graded 0–3) and respiratory depression; block-related complications (eg, post-dural puncture headache, paresthesia, visceral injury, pneumothorax, hematoma, or local anesthetic toxicity); surgical bleeding, intraoperative fluid volume, and vasopressor requirement; comorbidities (Charlson Comorbidity Index); 30-day postoperative complications (Clavien–Dindo classification); and overall morbidity (Comprehensive Complication Index).

Non-inferiority design

The study primarily aimed to compare the analgesic efficacy of the anterior subcostal QL block and ITM by assessing the median differences in 24 hours cumulative morphine consumption. A non-inferiority framework was applied, testing the non-inferiority hypothesis with a one-sided t-test at a 2.5% significance level.[19] The upper bound of the 95% CI for the difference (anterior subcostal QL block – ITM) was compared against the prespecified non-inferiority margin (Δ). If the upper

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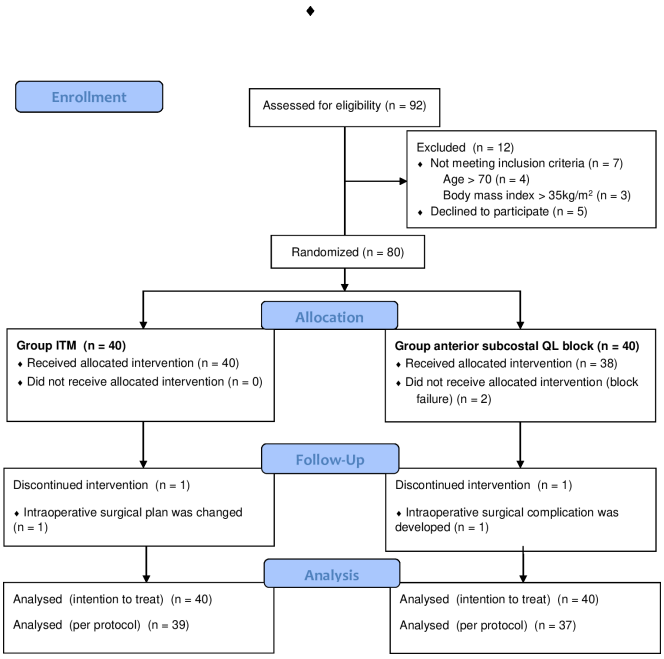

Figure 2 Consolidated Standards of Reporting Trials flow diagram of participants. ITM, intrathecal morphine; QL, quadratus lumborum.

bound remained within this margin, the anterior subcostal QL block was considered non-inferior to the ITM.

Non-inferiority margin

We selected the non-inferiority margin (Δ) using the fixed margin approach, informed by previous meta-analyses, a placebo-controlled study, and expert opinion, in accordance with FDA guidance and ICH E10 recommendations.[20 21] Based on prior findings, the 24 hours cumulative mean intravenous MME consumption differed between the control (15±8.8 mg) and ITM (2±6.8 mg) groups by 13 mg (95% CI 10.53 to 15.47). Our non-inferiority margin was set at approximately 50% of the lower bound of this CI, yielding 5.27 mg. A pooled SD of 7.86 mg was assumed for morphine consumption distribution.[22 23]

Statistical analysis

Sample size calculations

We determined that 36 participants per group would provide 80% power (β=0.20) to demonstrate the non-inferiority of the anterior subcostal QL block to ITM on the primary outcome, with a 2.5% significance level. The non-inferiority margin was 5.27 mg (favoring a lower mean), assuming a true mean difference of zero and pooled SD of 7.86. Allowing for 10% attrition, 40 participants per group (80 in total) were planned. The sample size and power calculations were performed using PASS 21 for a non-inferiority test of two means.

Data analyses

Data analyses were conducted using IBM SPSS Statistics (V.27.0, IBM Corp., Armonk, NY, USA). The Shapiro–Wilk test was used to assess the normality of continuous variables. Non-normally distributed variables were compared using the Mann–Whitney U test and are presented as medians (IQR), while normally distributed variables were compared using two-tailed Student's t-tests and are presented as means±SD. Categorical variables were analyzed using the $\chi 2$ or Fisher's exact test and expressed as counts (%). Bonferroni correction was applied when necessary, with statistical significance set at $p<0.0167$.

All secondary outcomes were analyzed using two-sided tests, with significance set at $p<0.05$. Analyses were performed based on the intention-to-treat (ITT) population, including all randomized patients receiving any part of the intervention. A per-protocol analysis was conducted to determine the primary outcome. Owing to the skewed distribution, non-inferiority of

Table 1 Demographic and surgical characteristics in the groups

Outcomes	Group ITM (n=40)	Group anterior subcostal QL block (n=40)	P value
Median age (Q1, Q3) in years	59.5 (53.5, 65.5)	59 (45,67)	
Sex, n (%)			
Female	9 (22.5)	17 (42.5)	
Male	31 (77.5)	23 (57.5)	
Median BMI (Q1, Q3) in kg/m²	25.45 (23.9, 26.8)	25.97 (24.2, 27.9)	
ASA physical status, n (%)			
I	5 (12.5)	7 (17.5)	
II	25 (62.5)	32 (80)	
III	10 (25)	1 (2.5)	
Comorbidities, n (%)			
None		9 (22.5)	0.68
Cardiovascular system disorders*	9 (22.5)	12 (30)	
Endocrine system disorders†	4 (10)	3 (7.5)	
>1 system involvement	16 (40)	8 (20)	
Other	1 (2.5)	8 (20)	
Type of surgery, n (%)			
Partial	24 (60)	26 (65)	0.81
Radical	16 (40)	14 (35)	
Median duration of surgery (Q1, Q3) in min	150 (105,180)	127.5 (130, 187.5)	**0.04**
Median duration of pneumoperitoneum (Q1, Q3) in min	120 (96.2,166.2)	92.5 (72.7, 123.7)	**0.007**
Mean intraoperative remifentanil consumption (SD) in µg	873.5 (470.6)	912.5 (425.7)	0.69
Median vasopressor use (Q1, Q3) in µg	0 (0, 23)	2 (0, 22)	0.84
Median crystalloid consumption (Q1, Q3) in ml	2000 (1775, 2575)	2000 (1500, 2950)	0.56
Median colloid consumption (Q1, Q3) in ml	0 (0, 500)	0 (0,0)	0.5
Median blood loss (Q1, Q3) in ml	200 (100, 387.5)	175 (100, 400)	0.44
Median urine output (Q1, Q3) in ml	300 (200, 400)	215 (100, 387.5)	0.11

Note Continuous variables are presented as median (Q1, Q3). Categorical variables were presented as counts (%). Statistically significant differences are highlighted in bold (p<0.05).
*Hypertension, coronary artery disease.
†Type 2 diabetes, goiter.
ASA, American Society of Anesthesiologists; BMI, body mass index; ITM, intrathecal morphine; QL, quadratus lumborum.

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the primary outcome was assessed using the upper 95% CI of the median difference via the Hodges-Lehmann method, with a one-sided non-inferiority threshold of p<0.025.

Sensitivity analysis of the primary outcome

Owing to the skewed morphine consumption, a sensitivity analysis was performed using the bias-corrected bootstrap method with 10 000 resamples in the per-protocol population. This method generates robust CIs without relying on normality assumptions, ensuring an accurate and reliable estimation of the median difference.[24 25]

RESULTS

Study participants

Between October 2024 and April 2025, 92 patients were screened, 80 of whom consented to participate and were randomized into two study groups. Seventy-six patients completed the study per protocol (ITM group, 39; anterior subcostal QL block group, 37). Four patients were excluded from the per-protocol analysis due to two block failures, one conversion to open nephrectomy in the anterior subcostal QL block group, and one surgical complication (diaphragmatic perforation with pneumothorax) in the ITM group. All 80 patients were included in the ITT analysis (figure 2). Demographic, surgical, and hemodynamic characteristics were comparable between the groups, except for longer anesthesia and pneumoperitoneum duration in the ITM group (p<0.05) (table 1 and online supplemental appendix 3).

Primary outcome: first 24-h opioid consumption non-inferiority

The ITT analysis for the primary outcome showed a median difference in cumulative 24-h intravenous MME consumption of 9.5 mg (95% CI 7 to 12; p<0.001) between the anterior subcostal QL block and ITM groups. The per-protocol analysis yielded a similar median difference of 9 mg (95% CI 7 to 12 mg; p<0.001), confirming consistency. Notably, the upper bound of the 95% CI (12 mg) exceeded the non-inferiority margin (Δ = 5.27 mg), indicating that the anterior subcostal QL block did not meet the non-inferiority threshold. Consequently, non-inferiority was not achieved in either the ITT or per-protocol analyses (figure 3). Although 24-h intravenous MME consumption was significantly lower in the ITM group and the non-inferiority margin was not crossed, the study design did not permit a formal claim of superiority.

Individual comparisons of opioid consumption

In the ITT population, the median 24-h cumulative intravenous MME consumption was 15 mg (Q1–Q3: 10–22.75 mg) in the anterior subcostal QL block group and 6 mg (Q1–Q3: 4–7 mg) in the ITM group. In the per-protocol analysis, the anterior subcostal QL block group also consumed 15 mg (Q1–Q3: 10–21 mg) compared with 6 mg (Q1–Q3: 4–7 mg) in the ITM group. Significant differences were observed between the groups in both the ITT (p<0.001; z = 5.817) and per-protocol analyses (p<0.001; z=5.591). No differences were noted between the

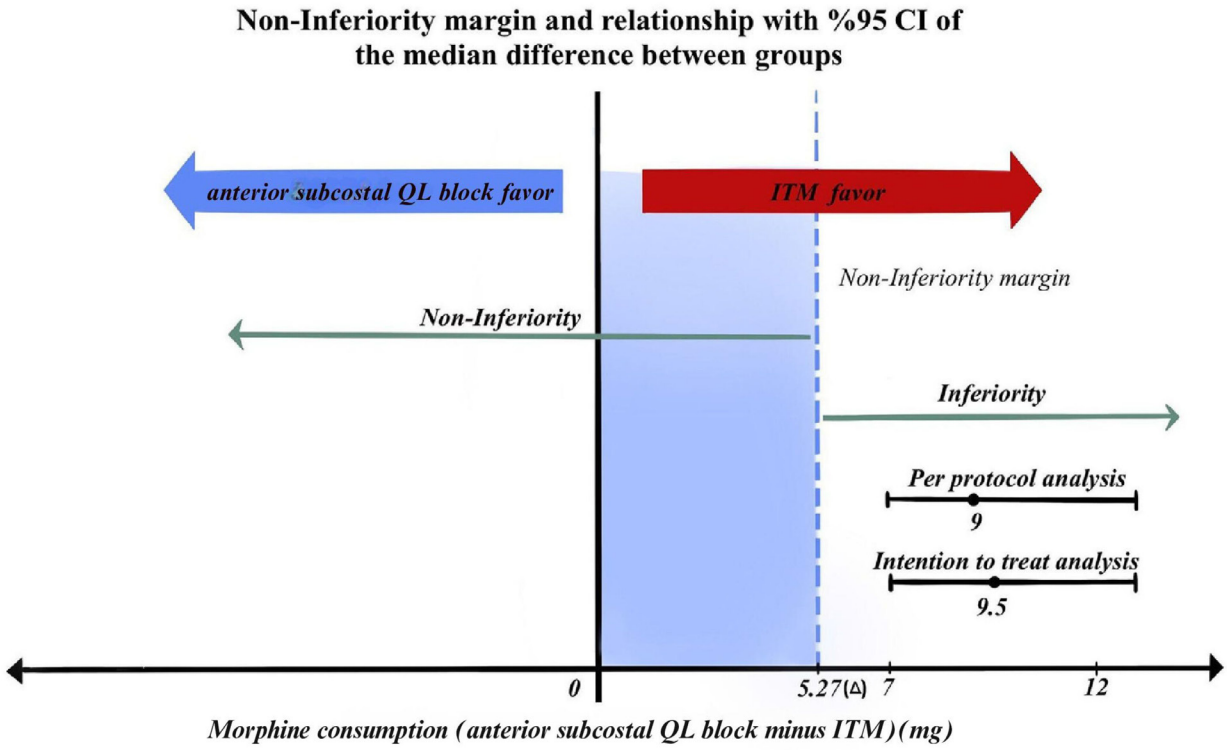

Figure 3 A non-inferiority diagram illustrating the observed difference in median cumulative 24-hour intravenous MME consumption between the anterior subcostal QL block and ITM groups. The dashed line represents the non-inferiority margin (Δ), while the error bar represents the 95% CI of the difference between the anterior subcostal QL block and ITM groups. ITM, intrathecal morphine; MME, morphine milligram equivalent; QL, quadratus lumborum.

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Table 2 Comparison of NRS pain scores at rest and with activity across time points and postoperative analgesic outcomes in the groups

Outcomes	Group ITM (n=40)	Group anterior subcostal QL block (n=40)	P value
Median NRS_{rest} (Q1, Q3)			
PACU	3 (2, 4)	5 (4, 7)	**<0.001**
3rd hour	2 (2, 2)	3 (2, 4)	**<0.001**
6th hour	2 (1, 2)	3 (2, 4)	**<0.001**
12th hour	1 (1, 2)	2 (2, 3.7)	**<0.001**
24th hour	1 (1, 2)	2 (1.25, 3)	**<0.001**
Median $NRS_{activity}$ (Q1, Q3)			
PACU	4 (3, 5)	6 (5, 8)	**<0.001**
3rd hour	3 (2, 3)	4 (3, 5)	**<0.001**
6th hour	2 (2, 3)	4 (3, 5)	**<0.001**
12th hour	2 (2, 3)	3 (3, 4)	**<0.001**
24th hour	2 (2, 2)	3 (2, 3)	**<0.001**
Median postoperative cumulative intravenous MME consumption first 12 hours (Q1, Q3) in mg	3 (2, 4)	11 (6, 15.5)	**<0.001**
Median postoperative cumulative intravenous MME consumption first 24 hours (Q1, Q3) in mg	6 (4, 7)	15 (10, 22.5)	**<0.001**
Median time to first PCA morphine (Q1, Q3) (min)	40 (15, 63.7)	30 (17.25, 3)	0.11
Rescue analgesics in 24 hours, n (%)	21 (52.5)	28 (70)	0.16

Note: Continuous variables are presented as medians (Q1, Q3). Categorical variables were presented as counts (%). Statistically significant differences are highlighted in bold (p<0.05).
ITM, intrathecal morphine; MME, morphine milligram equivalent; NRS, numeric rating scale; PACU, post-anesthesia care unit; PCA, patient-controlled analgesia; QL, quadratus lumborum.

groups regarding the time to the first opioid request or the need for rescue analgesia (table 2 and Online supplemental appendix 4).

Secondary outcomes

Safety data

Sedation scores were comparable across the groups, except for the third measurement, with no cases of postoperative respiratory depression. Antiemetic use rates were similar between the two groups (ITM: 0 vs anterior subcostal QL block: 3 patients, p=0.077). Pruritus was observed in three ITM and one anterior subcostal QL block (p=0.305), which resolved spontaneously in all cases. The ITM group reported postdural puncture headache (3 patients), paresthesia (3 patients), and motor block (1 patient), while 1 patient with anterior subcostal QL block experienced diaphragmatic perforation and pneumothorax due to surgery.

Pain scores and patient satisfaction

The NRS pain scores at rest and during activity were consistently lower in the ITM group than in the anterior subcostal QL block group (table 2). Patient satisfaction with analgesia was also higher in the ITM group at 24 h and at discharge ([median/Q1–Q3], ITM (108/102.25–113.5) vs anterior subcostal QL block (75.5/58–100) at 24 h; ITM (117/115–123.75) vs anterior subcostal QL block (94/77–110) at discharge, p<0.001) (table 3). These findings suggest a more favorable early postoperative experience with ITM.

DISCUSSION

Our study demonstrated that 24-h cumulative intravenous MME consumption was significantly lower in the ITM group than in the anterior subcostal QL block group, and non-inferiority of the anterior subcostal QL block to the ITM was not established. Patients with ITM consistently reported lower NRS pain scores, with no significant differences in sedation, antiemetic use, or pruritus between the groups. During the 30-day follow-up, no significant differences were found in the Clavien–Dindo classification or Comprehensive Complication Index scores, confirming similar safety and morbidity outcomes. Additionally, patients with ITM reported higher analgesia satisfaction at 24 h and upon discharge.

Although the anterior subcostal QL block targets both somatic and visceral analgesia, its limited efficacy in our study may stem from inconsistent cranial spread of the local anesthetic to the thoracic paravertebral space, resulting in insufficient blockade of thoracic sympathetic fibers or the paravertebral chain and reduced visceral pain control. The injection at the L1–L2 level targets the fascial plane between the QL and anterior thoracolumbar fascia, allowing cephalad diffusion through the lateral arcuate ligament; however, this pathway has shown variable results in cadaveric and imaging studies.[7–9] This pathway has been proposed to allow limited transition from the abdominal to

Table 3 Comparison of complication rates, comorbidity scores, and QoR-15 scores in the groups

Outcomes	Group ITM (n=40)	Group anterior subcostal QL block (n=40)	P value
Patients requiring antiemetics, n (%)	0 (0)	3 (7.5)	0.07
Patients with pruritus, n (%)	3 (7.5)	1 (2.5)	0.30
Median Comprehensive Complication Index Score (Q1,Q3)	8.7 (0, 8.7)	8.7 (0, 8.7)	0.47
Median Charlson Comorbidity Index Score (Q1,Q3)	0 (0, 2)	1 (0, 2)	0.15
Clavien–Dindo Classification, n (%)			
Grade 0	16 (40)	11 (27.5)	0.24*
Grade 1	23 (57.5)	29 (72.5)	
Grade 3	1 (2.5)	0 (0)	
Median sedation score (Q1,Q3)			
PACU	0 (0, 0)	0 (0, 0)	0.30
3rd hour	0 (0, 0)	0 (0, 0)	**0.04**
6th hour	0 (0, 0)	0 (0, 0)	0.30
12th hour	0 (0, 0)	0 (0, 0)	1
24th hour	0 (0, 0)	0 (0, 0)	1
Median QoR-15 Score (Q1,Q3)			
Preoperative	135 (132, 139.7)	137 (125, 140)	0.67
24th hour	108 (102.2, 113.5)	75.5 (58, 100.7)	**<0.001**
Discharge	117 (115, 123.7)	94 (77, 110)	**<0.001**

Note: Continuous variables are presented as median (Q1,Q3), and categorical variables are presented as counts (%). Statistically significant differences are highlighted in bold (p<0.05).
*For the Clavien-Dindo classification, a two-sided p<0.0167 was considered statistically significant after Bonferroni correction.
ITM, intrathecal morphine; PACU, post-anesthesia care unit; QoR-15, quality of recovery-15 score.

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the thoracic fascial planes, but reliable paravertebral distribution remains debated. Inadequate cephalad spread may restrict the involvement of thoracic sympathetic fibers and the paravertebral space, limiting visceral analgesia. Consequently, the anterior subcostal QL block may not match ITM in attenuating the visceral pain after laparoscopic nephrectomy. These findings underscore the need for imaging studies to clarify the anesthetic spread and optimize block techniques.

Elsharkawy's 2016 modification of the QL block involved shifting the injection site from the L4 to the L1–L2 level to promote cranial spread and achieve broader dermatomal coverage (T6–L2).[5 26] However, most clinical and cadaveric studies, including ours, reported consistent involvement of the T9–L1 dermatomes, with thoracic paravertebral spread observed infrequently and unpredictably.[6 27] The limited postoperative opioid-sparing effect of the anterior subcostal QL block may result from insufficient local anesthetic spread to the thoracic paravertebral region and limited sympathetic blockade, leading to inadequate visceral pain control. Moreover, the unilateral QL block may not effectively address contralateral trocar site pain, unlike ITM, which offers bilateral coverage via cerebrospinal diffusion.[2] Additionally, surgical trocar sites on the abdomen contralateral to the block could have contributed. Unlike unilateral QL block, ITM provides reliable bilateral coverage, which may contribute to its superior analgesic effect.[28 29] This approach may enhance patient comfort and minimize complication risks; however, the bilateral block may further decrease morphine requirements. This mechanism also explains the bilateral and rostral analgesic coverage achieved with ITM, particularly beneficial in laparoscopic surgeries.[2]

Following ITM administration, side effects, such as sedation, nausea, vomiting, pruritus, and urinary retention, may occur, with respiratory depression as a particularly limiting adverse effect. However, these effects are dose-dependent, and low doses of ITM (<200 mcg) without potentiating drugs (eg, benzodiazepines) are not associated with a higher risk of respiratory depression than systemic opioid therapies in clinical use.[2] In our study, safety outcomes showed no significant differences between the ITM and anterior subcostal QL block groups regarding sedation scores and postoperative complications, with no cases of respiratory depression in either group. Antiemetic use and pruritus rates were comparable between the groups. Although mild complications, such as postdural puncture headache, paresthesia, and motor block, were more frequent in the ITM group, they resolved spontaneously. The Clavien–Dindo classification and comprehensive complication index indicated no severe complications, and the low Charlson Comorbidity Index scores in both groups further support the safety and tolerability of both methods.

This variability in local anesthetic spread highlights the need for imaging-based studies to confirm thoracic paravertebral involvement and optimize the anterior subcostal QL block technique. Future studies should also aim to enhance visceral analgesia through improved block design, evaluate the impact of bilateral applications, and define appropriate non-inferiority margins using larger cohorts in multimodal analgesia strategies for laparoscopic surgery.

Limitations

This study had several limitations. First, the anterior subcostal QL block was performed unilaterally, and some trocar entry sites were located contralaterally. This may have resulted in incomplete somatic coverage, and bilateral application may have further reduced opioid consumption and improved analgesic outcomes. Second, the short follow-up period limited the evaluation of chronic pain and recovery.[2] Third, while opioid consumption served as the primary outcome, secondary measures such as functional recovery and patient satisfaction may not have been comprehensively assessed. Fourth, the limitation of this study was the lack of blinding of the proceduralist and participants. Another limitation, although remifentanil-related opioid-induced hyperalgesia is documented in the literature, similar intraoperative doses and standardized multimodal analgesia in our study suggested it did not significantly affect postoperative pain or opioid consumption between the groups.[30] Finally, despite the sample size being powered for primary outcomes, it may have been insufficient to detect minor differences in secondary outcomes, such as side effects or complications.

CONCLUSIONS

This study found that ITM resulted in significantly lower 24-h postoperative opioid consumption than unilateral anterior subcostal QL block in patients undergoing laparoscopic nephrectomy. However, as the trial was designed to assess non-inferiority rather than formally test for a difference, these results should be interpreted as failing to demonstrate the non-inferiority of the anterior subcostal QL block. Both interventions demonstrated comparable safety profiles, with no significant adverse events observed.

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Acknowledgements We would like to thank Editage [http://www.editage.com] for editing and reviewing this manuscript for the English language. During the preparation and editing of this manuscript, ChatGPT was utilized to assist with language refinement and clarity improvements.

Contributors CK, BD, ET, and YBU conceptualized the manuscript. BD, CK, ET, and ANK collected data and wrote the first draft. YBU, ANK, HC, and ADC conducted literature review and assisted in data analysis. CK, ET, HC, and BD wrote the final version of the manuscript with critical revisions from HE, ADC, and YBU. BD is the guarantor of this study. All authors contributed significantly to the study's design, conduct, and analysis. All authors reviewed and approved the final version of the manuscript for publication. During the preparation and editing of this manuscript, ChatGPT was utilized to assist with language refinement and clarity improvements.

Funding This study was supported by the Commission Presidency of Scientific Research Projects of Ondokuz Mayis University, Samsun, Turkiye, under project number BAP06-2024-4941. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Competing interests Hesham Elsharkawy: consultant for SPR Therapeutics, Curonix. Stock options Neuronoff, Clinical and Translational Science Collaborative (CTSC) of Cleveland, which is funded by the National Institutes of Health (NIH), National Center for Advancing Translational Science (NCATS), and Clinical and Translational Science Award (CTSA) grant, UL1TR002548. None of these is related to this work.

Patient consent for publication Consent obtained directly from patient(s).

Ethics approval This study involved human participants and was approved by the Ethics Committee of Ondokuz Mayıs University Faculty of Medicine (Approval No. 2023/212). Participants gave informed consent to participate in the study before taking part.

Provenance and peer review Not commissioned; externally peer-reviewed.

Data availability statement Data are available upon reasonable request. Data are available upon reasonable request. All data generated or analyzed during this study are included in this article and its supplementary material files. Further inquiries can be directed to the corresponding author.

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